Digital Transformation Vision

MES & AI■Native Manufacturing

Prepared for: Solar Manufacturer | 02 May 2026

Vision Statement

To become a data■driven, AI■native solar manufacturer where every cell, module, and pallet is tracked in real time, quality is predicted before it fails, and the factory optimises itself autonomously. This transformation will turn 300 MW annually of production into a fully visible, ALMM■compliant, and highly efficient digital operation.

Business Drivers

- Real■time visibility into production OEE and quality (FPY)
- Complete material genealogy for ALMM compliance and warranty reduction
- Intelligent scheduling (PODISE) to maximise line utilisation and on■time delivery
- Predictive maintenance to cut unplanned downtime by >25%
- AI■powered quality inspection to reduce manual checks and warranty claims

Expected Benefits

Benefit	Impact
	+8■12%
First Pass Yield	>98%
	30■40%
Warranty Claims Reduction	20■30%
	80% reduction
Operator Efficiency	50%+ reduction in manual data entry